

JHA SAKET SUNIL

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EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal

Aug 2023 – May 2027

Bachelor of Technology, Civil Engineering

CGPA: 8.63

Minor in Information Technology

CGPA: 9.75

EXPERIENCE

AI Engineer Intern

May 2025 – July 2025

Kalppo

- Developed a scalable pipeline using Google Cloud Vision, structured content generation, and fuzzy sequence matching to automate student workbook evaluation, reducing manual assessment time by 80%.
- Containerized a FastAPI-based REST service for input-driven STEM question and SMILES ↗ code generation, featuring a LaTeX-to-image rendering pipeline with Supabase integration for media storage.
- Collaborated on building an AI-driven microservice for automated question extraction from PDFs using Mistral OCR, LangChain, and Gemini for structured parsing, with PostgreSQL for metadata storage.

PROJECTS

🔗 Paragon || C++, CMake, OpenMP, pybind11, pip

Jan 2026 - Present

- Developed PARAGON, a high-performance parallel graph processing library featuring a C++ computational core with Python bindings, published as a Python package while maintaining performance critical logic in C++.
- Built using C++, Python, CMake, OpenMP, and pybind11, implementing parallel graph algorithms including PageRank, SSSP, and Triangle Counting with optimized data structures and thread level parallelism.
- Benchmarked against sequential implementations, achieving up to 80× speedup in SSSP, 3.4× in Triangle Counting, and 2.5× in PageRank on multicore CPUs.

🔗 FileForge || FastAPI, PostgreSQL, MinIO, Docker, Alembic

May 2024 - July 2025

- Designed and developed a distributed file storage system supporting nested folders, file and folder uploads, ZIP extraction, and a personalized favorites system with secure, user-level access.
- Built secure authentication and session management workflows using JWT and OAuth2.
- Integrated MinIO with optimized PostgreSQL indexing, achieving 40% faster file metadata queries.

🔗 ATLAS || YOLO, Ultralytics, ByteTrack, Pydantic

March 2024 - Dec 2025

- Developed ATLAS (Adaptive Traffic Light Allocation System) using YOLOv8 and ByteTrack for real time vehicle detection, tracking, and classification with Region Mask ↗ and frame-based confirmation filtering.
- Implemented DBWSA ↗ to dynamically assign green signal time based on vehicle density and vehicle type.
- Achieved 25% average and 50% peak efficiency gains over fixed signals across 500+ simulated traffic scenarios.

TECHNICAL SKILLS

Languages & Databases

Python, C/C++, SQL, PostgreSQL, MySQL, Redis, MinIO, ChromaDB

Developer Tools

Git, Docker, Bash, Google Cloud, Kubernetes, MLflow

Frameworks

FastAPI, LangChain, PyTorch, TensorFlow, SQLAlchemy, Hugging Face

Libraries

Scikit-learn, Pydantic, Pandas, Alembic, SpaCy, OpenCV, Celery

EXTRACURRICULAR

Web Club, NITK – Executive Member

Oct 2024 - Present

Organized Silicon Maze, Web Club's flagship event, successfully coordinating 500+ online participants.
Conducted session on Mamba SSM ↗, highlighting its architecture and advantages for efficient sequence modeling.

IE, NITK – Executive Member

Sept 2024 - Present

Assisted in building a YOLOv5 based fire-fighting robot at IE NITK, achieving 86% mAP@0.95

ACHIEVEMENTS

- Winner, Kaggle x Skill Assessment Machine Learning Competition, Rank 1 among 1,846 participants.
- Competitive Programming: LeetCode (Contest Rating: 1667), Codeforces (Contest Rating: 1111)